



Constellation®

LaSalle County Station

2601 North 21st Road
Marseilles, IL 61341
815-415-2000 Telephone

10 CFR 50.73

RA23-021

May 2, 2023

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

LaSalle County Station, 2
Renewed Facility Operating License No. NPF-11
NRC Docket No. 50-374

Subject: Licensee Event Report 2023-003-00, Automatic Actuation of Reactor Protection System (RPS) During Restoration from Hydrostatic Test Conditions

In accordance with 10 CFR 50.73(a)(2)(iv)(A), Constellation Energy Generation, LLC (CEG) is submitting Licensee Event Report (LER) Number 2023-003-00 for LaSalle County Station, Unit 2.

There are no regulatory commitments in this letter. Should you have any questions concerning this report, please contact Mr. Dan Mearhoff, Regulatory Assurance Manager, at (815) 415-2800.

Respectfully,

John Van Fleet Jr
Acting Site Vice President
LaSalle County Station

Enclosure: Licensee Event Report

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

LaSalle County Station, Unit 2

☒ 050
☐ 052

2. Docket Number

374

3. Page

1 OF 3

4. Title

Automatic Actuation of Reactor Protection System (RPS)

5. Event Date

6. LER Number

7. Report Date

8. Other Facilities Involved

Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
03	04	2023	2023	- 03 -	00	5	02	2023	Facility Name	☐ 052	Docket Number

9. Operating Mode

4

10. Power Level

0 Percent

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

CJ Smith, Operations Director

Phone Number (Include area code)

(815)-415-2200

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected

☒ No☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month

Day

Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

At 0910 CST on March 4, 2023, with Unit 2 in Mode 4 at 0 percent power, an actuation of a reactor scram on low charging water header pressure occurred during restoration from ASME Class 1 System Leakage Test conditions. All control rods were already fully inserted prior to the receipt of the scram signal.

The scram occurred due to a valid signal and all systems responded as expected. Low charging water header pressure is not required to be operable in Mode 4 but is active if not bypassed. The scram signal should have remained bypassed per station procedures.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME

LaSalle County Station, Unit 2

☒ 050☐ 052**2. DOCKET NUMBER**

374

3. LER NUMBER

YEAR	SEQUENTIAL NUMBER	REV NO.
2023	- 003	- 00

NARRATIVE

LaSalle County Station Unit 2 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power. The affected system was the Reactor Protection System (RPS).

RPS initiates a rapid insertion of all the control rods (scram) when monitored variables exceed their pre-established limits, neutron flux instrumentation becomes inoperable, or a manual scram signal is inserted by the operator. In addition, the setpoints, power sources, and controls and instrumentation are arranged in such a manner as to preclude spurious scrams.

Condition Prior to Event

Unit(s): 2

Date: March 4, 2023

Time: 0910 CST

Reactor Mode(s): 4

Mode(s) Name: Cold Shutdown

Power Level: 0%

Description

Following completion of required ASME Class 1 System Leakage Testing, actions were taken to commence reactor depressurization. As governed by the testing procedure, the Control Rod Drive (CRD) Charging Water Stop Valve was closed. The Low Charging Water Header Pressure scram signal was verified to be bypassed at the start of the Hydrostatic Test. The scram signal was then restored during the transition into scram time testing. With the CRD Charging water Stop valve closed and the Low Charging Water Header Pressure scram signal not bypassed, reactor depressurization resulted in an RPS Scram on Low Charging Water Header Pressure.

At the time of the Scram, the unit was in Mode 4 with all rods inserted. The Low Charging Water Scram Signal is not required in Mode 4 for Rod Cycling and the Scram signal should have remained bypassed, in accordance with station procedures.

Reportability and Safety Analysis

The Low Charging Water Pressure Scram is only required to be enabled in Mode 2 and Mode 5 to ensure that a reactor scram is initiated before Scram capability is compromised during low reactor pressure operations. At the time of the event the reactor was in Mode 4 with all rods inserted. Therefore, there was no increased risk to the public as a result of this event.

The event is reportable under 10 CFR 50.72(b)(3)(iv)(A) as an event or condition that results in the valid actuation of the Unit 2 RPS system. An ENS report was made to the NRC at 1642 on March 4, 2023 (EN# 56389).

Safety System Functional Failure Review

The system equipment responses did not result in a safety system functional failure (SSFF) as defined in accordance with NEI 99-02, "Regulatory Assessment Performance Indicator Guideline." The RPS logic safety function was satisfied by its actuation for designed inputs.



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	☐ 052		<table border="1"><tr><th>YEAR</th><th>SEQUENTIAL NUMBER</th><th>REV NO.</th></tr><tr><td>2023</td><td>- 003</td><td>- 00</td></tr></table>	YEAR	SEQUENTIAL NUMBER	REV NO.	2023
YEAR	SEQUENTIAL NUMBER	REV NO.					
2023	- 003	- 00					

NARRATIVE

Corrective Actions

The personnel involved in the event were coached in accordance with Constellation processes.

Station procedures will be revised include steps to verify the Low Charging Water Header Pressure Scram is bypassed prior to closing the CRD charging water stop valve.